

From 5 Million Dirty Rows to One Trusted Catalog

A consumer-packaged-goods data story — 5,014,409 catalog rows modelled on Libstar's public brand portfolio, cleaned by Azure Data Factory, served by Synapse serverless SQL, scored with machine learning, and reconciled to the rand across Qlik Sense, Excel and this page.

Case-study note: Libstar is a real JSE-listed CPG group, but every record here is synthetic — generated (seeded, reproducible) on Libstar's *public* brand/category model to demonstrate the pipeline without using any private data. Not affiliated with Libstar Holdings.

398 650

raw rows generated (incl. planted duplicates)

334 585

clean unique products after ADF

56 835

rows quarantined — each with a reason

R1.17tn

trailing 12-month revenue (synthetic)

0.71

price-model R^2 · MAE R12.71

1 · More than a product list

CPG catalogs rot quietly. Prices arrive as "R 1,234.56" from one system and "ZAR1234" from another; weights get captured in grams instead of kilograms; brands are typed by hand; category codes from a previous org structure ("Groceries", "Snacks and confectionery") linger years after a restructure. None of it looks urgent — until mispriced SKUs erode margin, duplicate listings distort demand forecasts, and rollup reporting breaks exactly when a range review needs it.

This project demonstrates the governed fix: a pipeline that **quantifies** the mess instead of silently filtering it, resolves it with auditable rules, and publishes one set of numbers that every downstream consumer — the live Qlik Sense app, the Excel pack, the ML models, this page — reconciles to **exactly**.

2 · The pipeline: dirty in, trusted out

```
generate_dirty_products.py (5.1M rows, seeded mess)
  | gzip CSV
  ▼
ADLS Gen2 raw/ → reference/ (brand map + dims)
  |
  ▼
Azure Data Factory · mapping data flow (8 vCores, ~6 min, ≈ R10/run)
  price/date/unit/boolean normalisation · brand lookup resolves misspelled
  brands AND legacy category names · dedupe (latest wins) · quarantine
  | snappy parquet
  ▼
```

ADLS Gen2 curated/ → Synapse serverless SQL (rpt.* views, \$0 idle)

- Qlik Sense Cloud app (5 sheets, built via qlik-cli)
- scikit-learn (pricing · IsolationForest · KMeans)
- Excel report (formula-driven, reconciled)
- Ebook + this data story

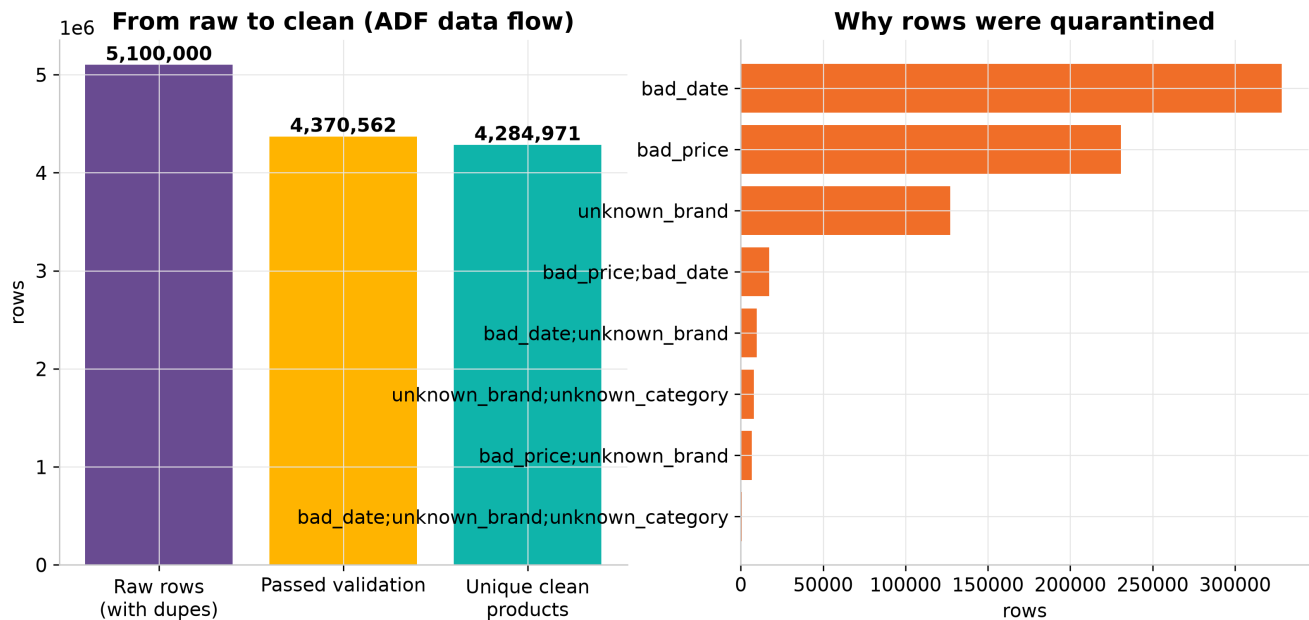


Figure 1 — 5,014,409 deduplicated rows in, 4,284,971 unique clean products out. Dates and prices are the biggest offenders.

3 · What the shelf looks like

Libstar's simplified operating model has two product groups. Ambient categories carry **77% of revenue** (R904.1bn of R1,170.3bn), led by Spreads, Meal ingredients and Wet condiments; perishables — dairy, convenience meals, value-added meats, baby, fresh mushrooms — hold the rest. Margins cluster tightly around the 35% portfolio average, so revenue mix, not margin spread, is what moves the total.

Revenue by category — Ambient does the heavy lifting

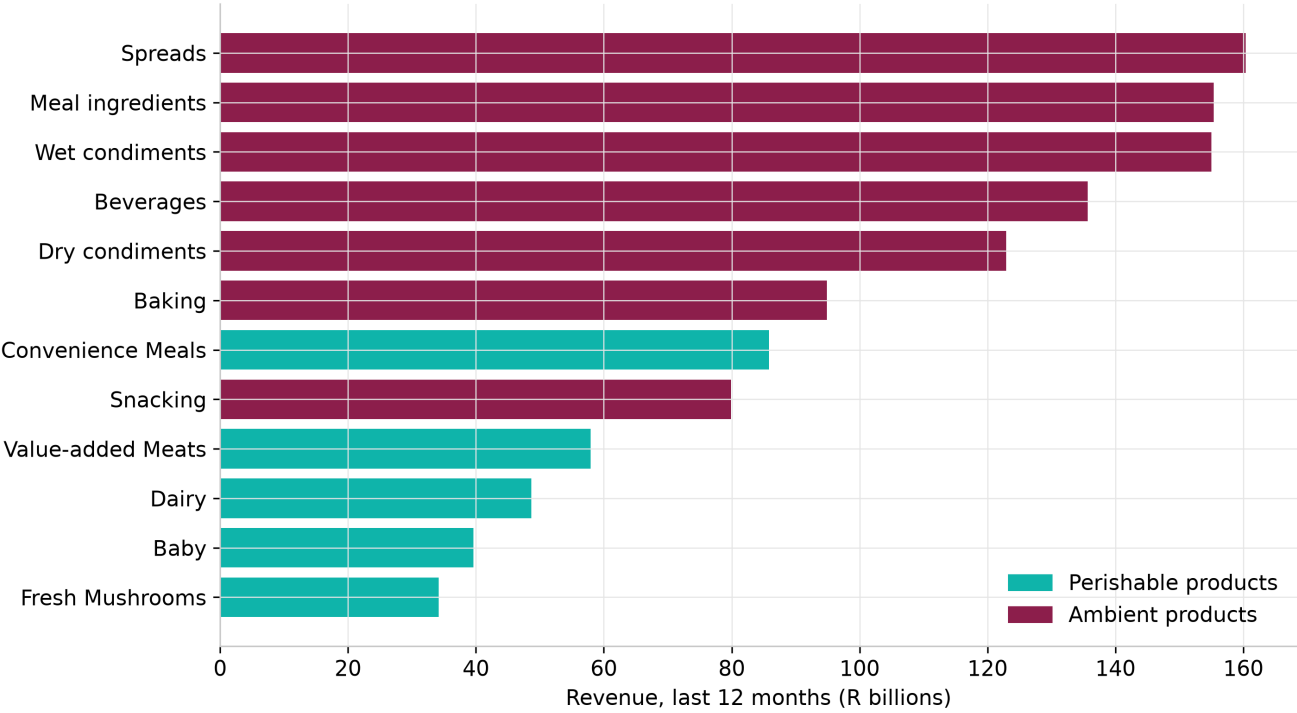


Figure 2 — Revenue by category, coloured by product group.

Margin by category

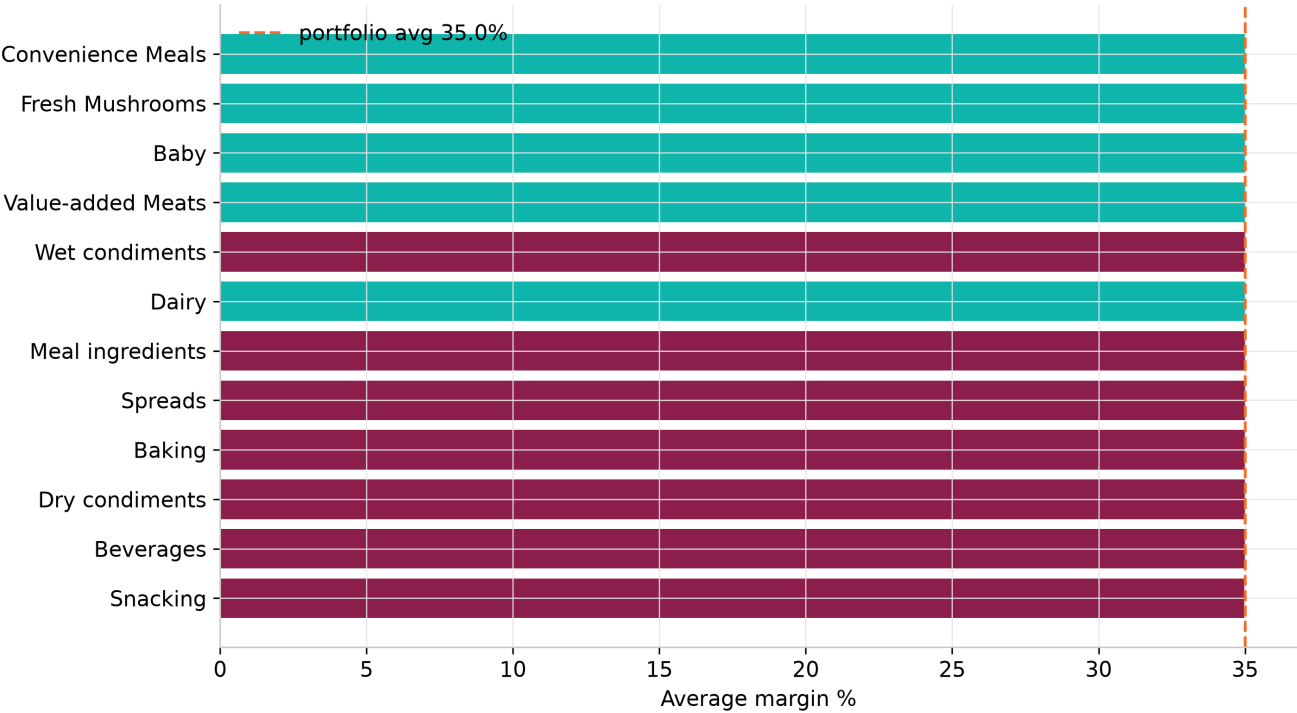


Figure 3 — Margin by category vs the portfolio average.

Category scorecard (full population)

Category	Group	SKUs	Revenue 12m	Avg margin	Avg price
Spreads	Ambient products	526,813	R160,395,514,140	35.0%	R87.63
Meal ingredients	Ambient products	554,781	R155,347,925,715	35.0%	R74.20
Wet condiments	Ambient products	580,617	R154,977,096,220	35.0%	R68.20
Beverages	Ambient products	428,482	R135,697,476,592	35.0%	R94.98
Dry condiments	Ambient products	452,143	R122,903,791,560	35.0%	R69.79
Baking	Ambient products	427,832	R94,861,638,385	35.0%	R46.87
Convenience Meals	Perishable products	295,388	R85,861,274,105	35.0%	R79.71
Snacking	Ambient products	311,271	R79,899,697,159	35.0%	R62.37
Value-added Meats	Perishable products	176,523	R57,959,630,790	35.0%	R102.24
Dairy	Perishable products	194,459	R48,642,328,237	35.0%	R59.31
Baby	Perishable products	168,555	R39,589,869,572	35.0%	R52.17
Fresh Mushrooms	Perishable products	168,107	R34,146,064,405	35.0%	R39.17

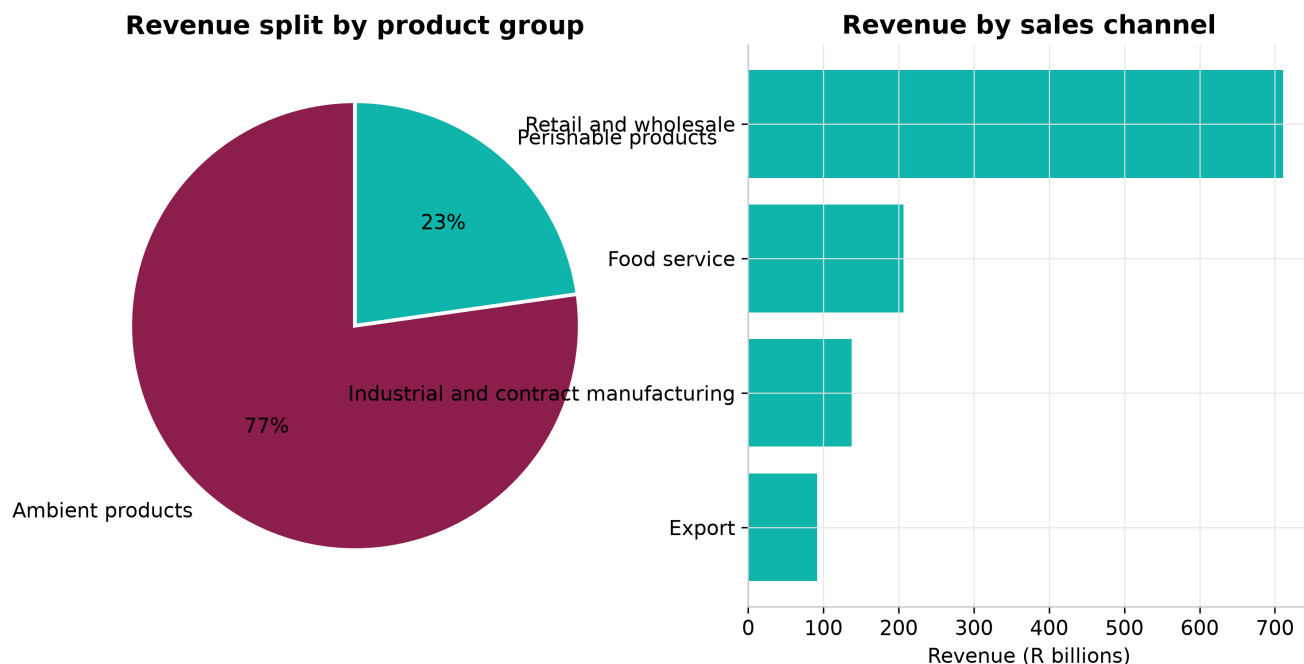


Figure 4 — Product-group split and the four routes to market.

4 • Brand power: three ways to win

Three brand solutions compete on the same shelf: Libstar's own brands (Lancewood, Denny Mushrooms, Cape Herb & Spice...), Principal brands represented in South Africa (Bonne Maman, Kikkoman, Tabasco, Maille), and private label for retailers. Principal brands price ~35% above comparable own-brand products; private label sits ~20% below — yet "Retailer Brands" is the single biggest revenue line (R103.7bn) because it spans all twelve categories.

Top 15 brands by revenue

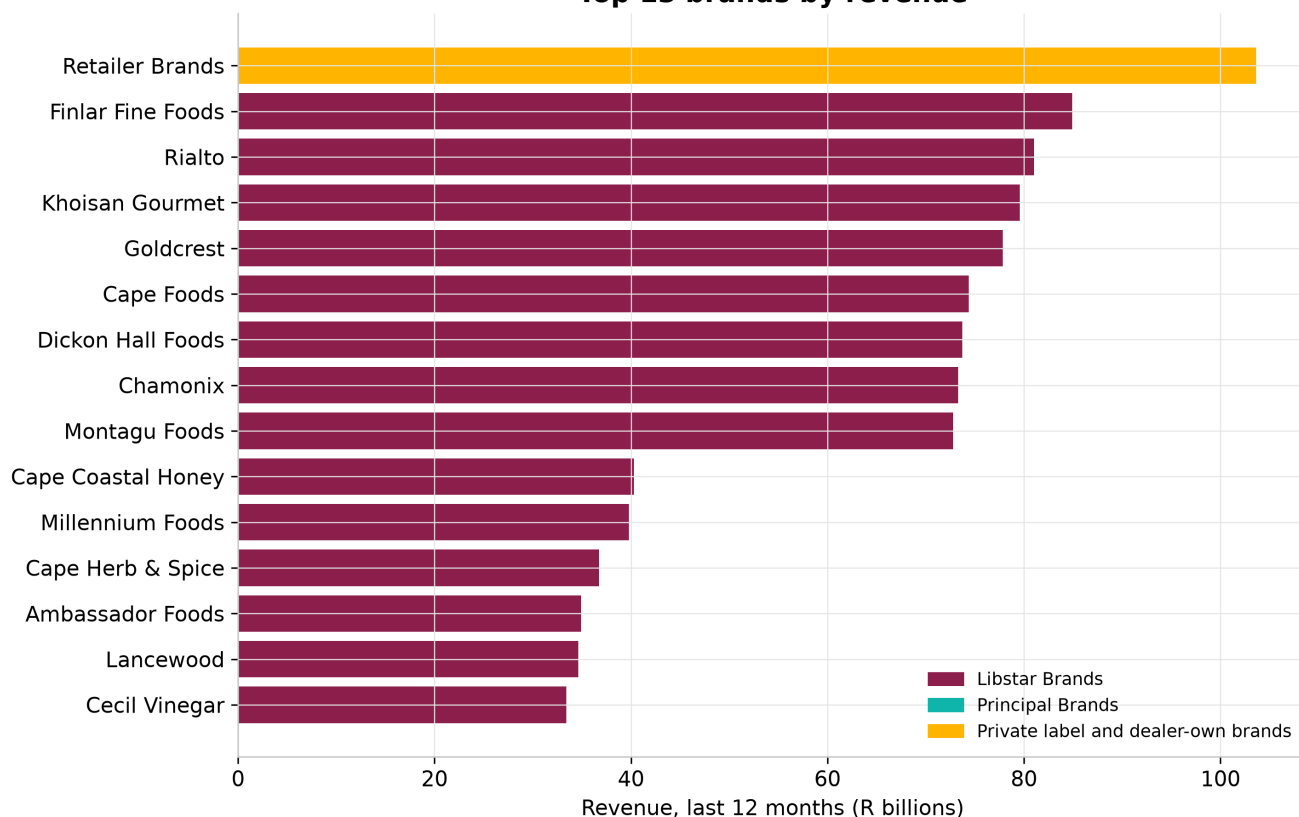


Figure 5 — Top 15 brands by revenue, coloured by brand solution.

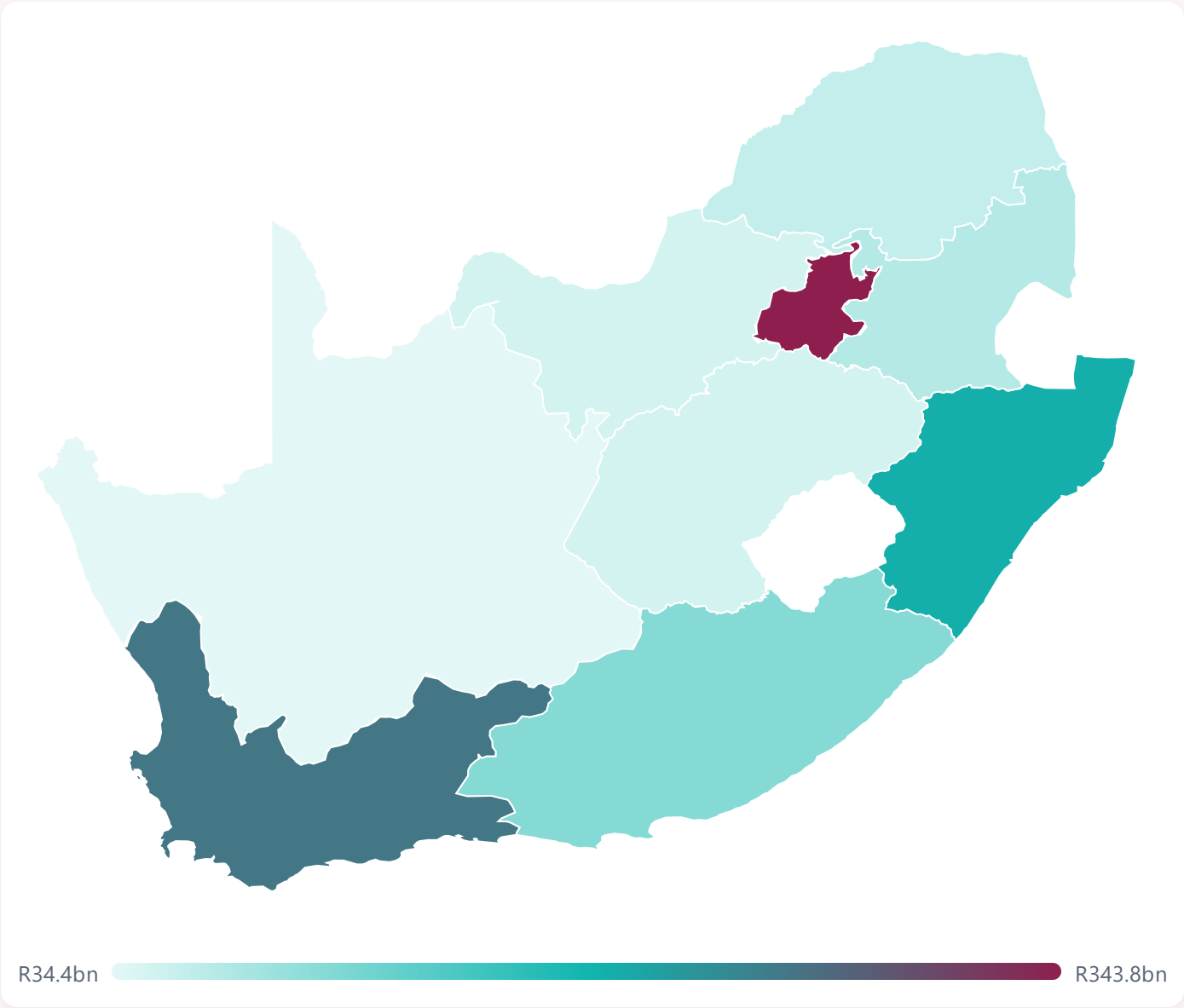
Top 10 brands

Brand	Brand solution	SKUs	Revenue 12m	Avg margin
Retailer Brands	Private label and dealer-own brands	430,236	R103,650,387,104	35.0%
Finlar Fine Foods	Libstar Brands	268,984	R84,928,535,830	35.0%
Rialto	Libstar Brands	269,175	R81,006,558,210	35.0%
Khoisan Gourmet	Libstar Brands	267,412	R79,563,401,192	35.0%
Goldcrest	Libstar Brands	268,612	R77,820,300,503	35.0%
Cape Foods	Libstar Brands	268,381	R74,364,367,070	35.0%
Dickon Hall Foods	Libstar Brands	269,141	R73,699,851,203	35.0%
Chamonix	Libstar Brands	269,004	R73,286,135,131	35.0%

Montagu Foods	Libstar Brands	269,449	R72,784,041,446	35.0%
Cape Coastal Honey	Libstar Brands	134,815	R40,291,885,412	35.0%

5 • Where South Africa buys

Hover (or tap) a province. Gauteng leads (R343.8bn), then Western Cape and KwaZulu-Natal — sales concentrate in the five provinces where the business has production or distribution footprints.



Hover a province

12-month revenue



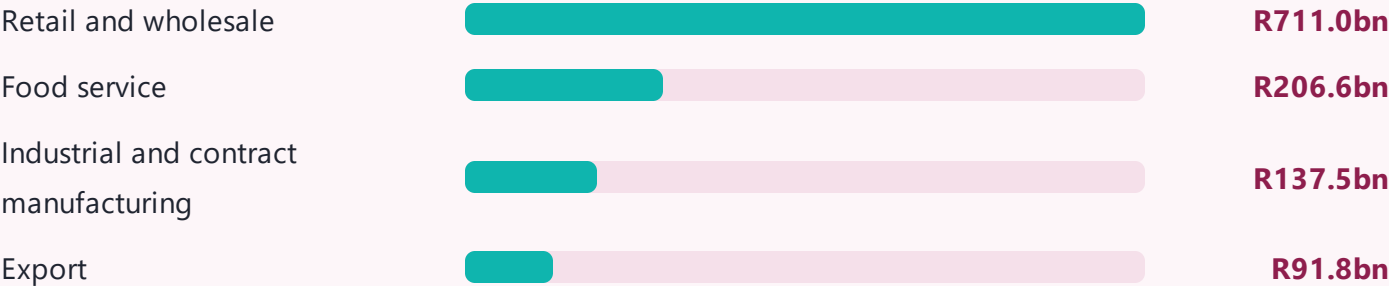
Products stocked



Average margin



Routes to market



6 · Data quality — where the numbers leak

The Data Factory flow rejected **729,438 rows (14.5%)** — and can say exactly why, row by row. Because reject reasons ship into the same reporting layer as the clean data, data quality becomes a trend a team can manage down, not an invisible loss.

Reject reason	Rows
bad_date	328,279
bad_price	230,750
unknown_brand	127,137
bad_price + bad_date	17,200

bad_date + unknown_brand	9,759
unknown_brand + unknown_category	8,039
bad_price + unknown_brand	6,615
bad_date + unknown_brand + unknown_category	657
bad_price + bad_date + unknown_brand	516
bad_price + unknown_brand + unknown_category	450
bad_price + bad_date + unknown_brand + unknown_category	36

Reconciliation guarantee: the Qlik app, the Excel report, the ebook and this page all read the same aggregate layer. Revenue is **R1,170,282,306,879** in all four — to the rand.

7 • Machine learning: pricing & anomalies

Pricing has learnable structure. A gradient-boosted model predicts shelf price from category, brand solution, channel, weight and rating with $R^2 = 0.71$ and a mean absolute error of **R12.71** on a mean price of R71.69 — evidence that pricing follows the portfolio's architecture, exactly what a well-governed catalog should show.

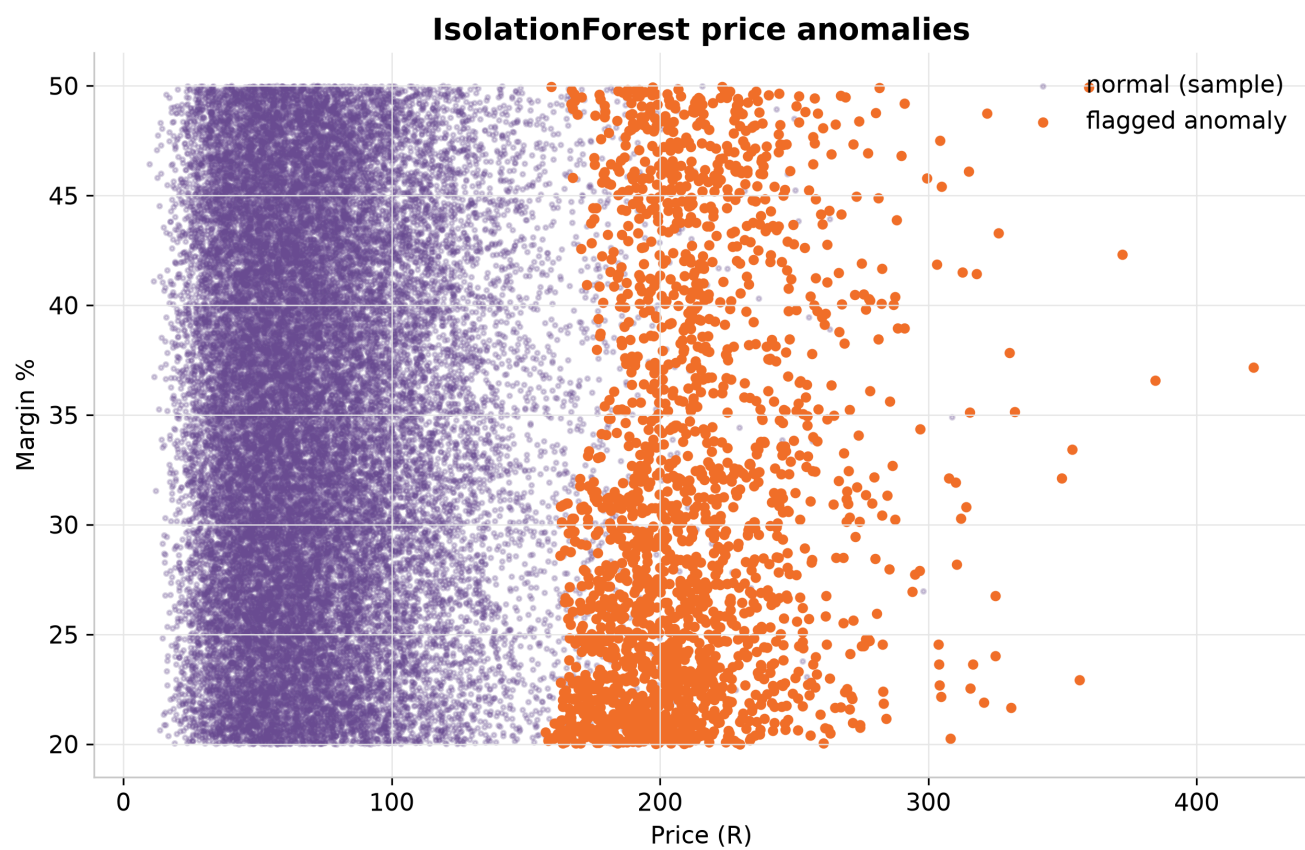


Figure 6 — IsolationForest anomalies vs the normal price–margin cloud.

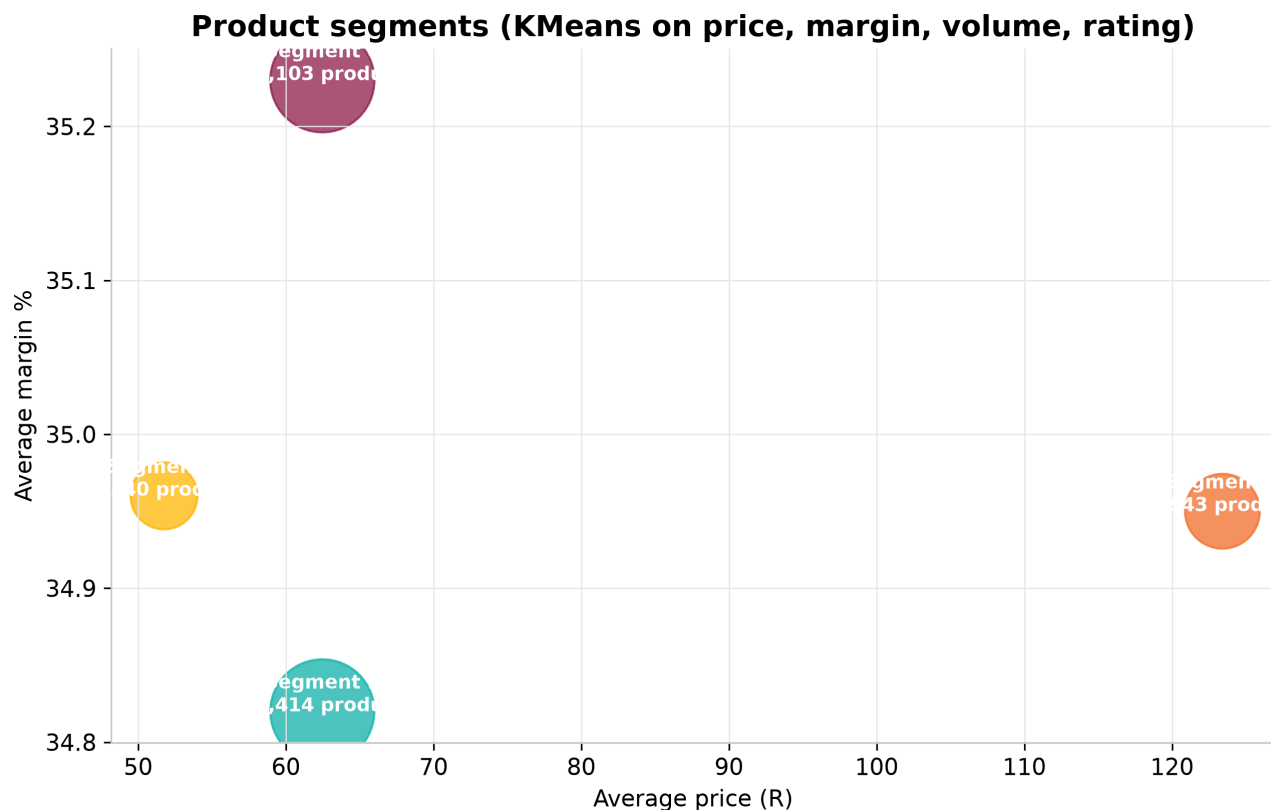


Figure 7 — Four KMeans segments: volume movers, premium niche, two mainstream clusters.

Why items get flagged, in plain language: the model scored one million products on price, cost, margin and weight *together*, and flagged the 2,000 whose combination is most unlike the rest — a heavyweight item priced like a small pack, an implausible margin, a luxury-priced commodity. These usually signal data-entry or pricing errors, not genuine premium products. **Recommendation:** route this list to category managers before the next price-list publication — correcting them before they reach shelves protects margin and avoids customer-facing pricing errors.

Top 10 to investigate first

Product	Category	Brand	Price	Cost	Margin	Score
Classic Beef Burger Patties 250ml	Value-added Meats	Finlar Fine Foods	R227.69	R181.82	20.1%	-0.0801

Premium Orange Marmalade 750g	Spreads	Bonne Maman	R260.87	R208.63	20.0%	-0.0791
Everyday Hot Chocolate 250ml	Beverages	Rialto	R315.01	R169.81	46.1%	-0.0760
Classic Strawberry Preserve 2kg	Spreads	Dickon Hall Foods	R193.94	R154.88	20.1%	-0.0749
Select Chicken Nuggets 250g	Value-added Meats	Finlar Fine Foods	R210.95	R168.58	20.1%	-0.0749
Everyday Chicken Nuggets 500ml	Value-added Meats	Finlar Fine Foods	R224.52	R179.44	20.1%	-0.0749
Traditional Mac & Cheese Meal 1kg	Convenience Meals	Millennium Foods	R219.97	R174.70	20.6%	-0.0748
Classic Instant Cappuccino 250ml	Beverages	Chamonix	R281.27	R155.00	44.9%	-0.0739
Organic Green Tea 500ml	Beverages	Chamonix	R203.88	R162.23	20.4%	-0.0732
Gourmet Apricot Jam 2kg	Spreads	Cape Coastal Honey	R236.00	R186.40	21.0%	-0.0724

8 · Azure in production

Everything runs on a free-trial credit by design — no always-on compute anywhere in the chain.

Resource	Purpose	Idle cost
stpargoparcels01	ADLS Gen2 — raw / curated zones	≈ \$0.03/month
adf-pargoparcels-za	Data Factory — cleaning data flow	\$0 (per-run billing)

`syn-pargoparcels`

Synapse serverless SQL endpoint

\$0 (\$5/TB scanned)

`pargo-trial-budget`

\$200 budget · email alerts at 50/75/90%

free

Kill-switch once all consumers are loaded: `az group delete --name rg-pargoparcels --yes`. The Qlik app (data loads in-app), the 36 MB .qvf export, the Excel pack, the ebook and this page all keep working after teardown.

9 · How it's built

Layer	Tools
Data generation	Python (pandas, numpy) — seeded synthetic catalog, reproducible mess
Storage	Azure Data Lake Storage Gen2 (raw / curated zones, Git LFS mirror in repo)
Transformation	Azure Data Factory mapping data flows
Serving	Azure Synapse Analytics serverless SQL
Machine learning	scikit-learn — HistGradientBoosting, IsolationForest, KMeans
BI	Qlik Sense Cloud — app, measures and 5 sheets built programmatically via qlik-cli
Reporting	openpyxl (Excel) · python-docx (ebook) · matplotlib (charts + SA map) · this page
Ops & cost	Azure CLI, PowerShell, Cost Management budget alerts

Rebuild from a clean clone:

```
git lfs pull
python scripts/build_reference_dims.py && python scripts/build_brand_map.py
python scripts/generate_dirty_products.py          # ~15 min
adf\deploy_adf.ps1 → run pl_clean_products → synapse\deploy_synapse.ps1
python ml/train_models.py && python scripts/make_charts.py
```

```
python scripts/build_excel_report.py && python scripts/build_ebook.py  
python scripts/build_data_story.py # this page
```

Libstar Product Analytics — synthetic portfolio project by Anthony Apollis · July 2026

Data modelled on Libstar's public brand/category structure (libstar.co.za) · Not affiliated with Libstar Holdings